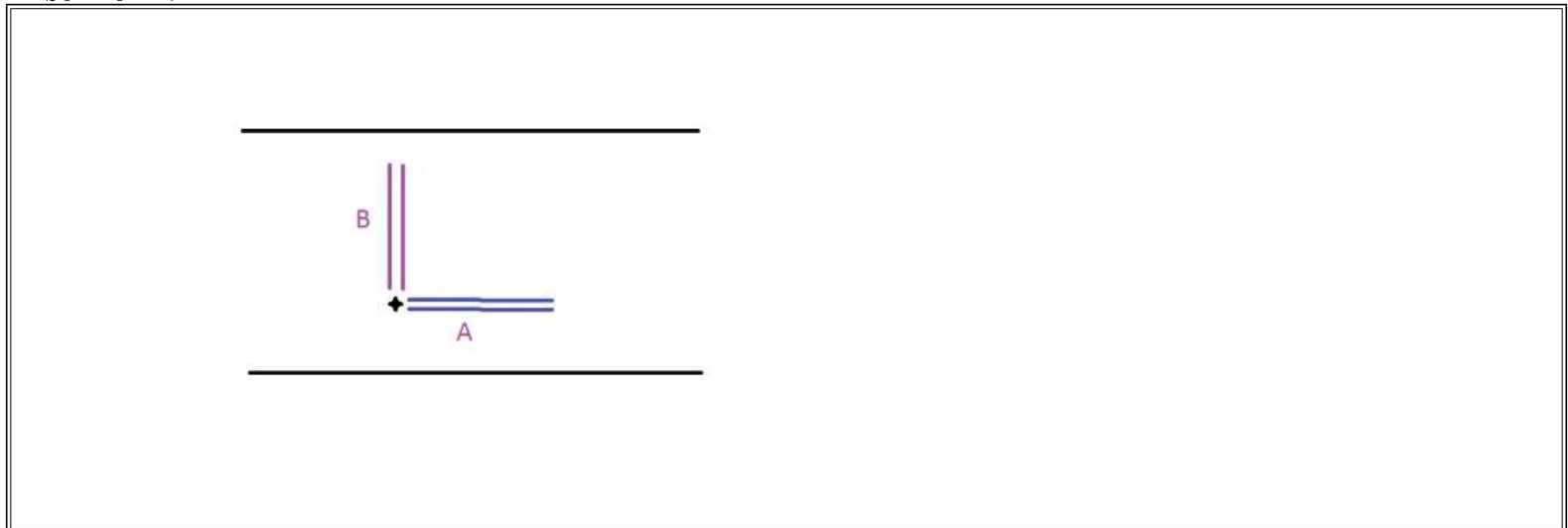


Question 1.7 : Two boats, A and B, move away from a buoy anchored at the middle of a river along the mutually perpendicular straight lines: the boat A along the river, and the boat B across the river. Having moved off an equal distance from the buoy the boats returned. Find the ratio of times of motion of boats τ_A/τ_B if the velocity of each boat with respect to water is $\eta = 1.2$ times greater than the stream velocity.

Solution: ->



Let the stream velocity be v_o , then the boat velocity with respect to water is ηv_o . Also let us assume that the equal distance be d .

Case A : Final velocity of boat A in forward journey (in the stream direction)

$$v = \eta v_o + v_o = (\eta + 1)v_o$$

$$\text{Time to reach the destination} = t_1 = \frac{d}{(\eta + 1)v_o} \dots\dots\dots(1)$$

Final velocity of boat A in the backward journey (opposite to the stream direction)

$$v = \eta v_o - v_o = (\eta - 1)v_o$$

$$\text{Time to reach back to the buoy} = t_2 = \frac{d}{(\eta - 1)v_o} \dots\dots\dots(2)$$

$$\tau_A = t_1 + t_2 = \frac{d}{(\eta + 1)v_o} + \frac{d}{(\eta - 1)v_o} = \frac{2\eta}{\eta^2 - 1} \frac{d}{v_o}$$

Case B : Here we proceed according to the shortest path case in Theory Problem 1.

In Forward Journey (Upwards in the figure)

$$t_1 = \frac{d}{\sqrt{\eta^2 - 1} v_o}$$

In Backwards Journey (Downwards in the figure)

$$t_2 = \frac{d}{\sqrt{\eta^2 - 1} v_o} = t_1$$

$$\tau_B = t_1 + t_2 = 2t_1 = \frac{2}{\sqrt{\eta^2 - 1}} \frac{d}{v_o}$$

$$\frac{\tau_A}{\tau_B} = \frac{\frac{2\eta}{\eta^2 - 1} \frac{d}{v_o}}{\frac{2}{\sqrt{\eta^2 - 1}} \frac{d}{v_o}} = \frac{\eta}{\sqrt{\eta^2 - 1}}$$

Substituting the value, we get $6/\sqrt{11}$ as the ratio. = 1.81 (approx.)